# THE COMPLETION OF A SUBMODULE, A MODULE AND ITS QUOTIENT FORM A SHORT EXACT SEQUENCE

THE STACKS PROJECT AUTHORS

*Editorial note.* All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file formal-spaces.tex; no statement or proof step was rewritten, reordered or removed.

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*Editorial note.* The proof span cites two same-chapter lemmas, delivered as navigation stubs.

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*Allowed standard prerequisites (editorial).*

- The closure of the image of N in the completion of M, and exactness at the left. Same chapter, Lemma lemma-closure, invoked at proof lines 1012-1013. Gives the exactness of 0 -> N^ -> M^ -> (M/N)^ and statement (2).
- Completeness of a quotient by a closed submodule and openness of the quotient map. Same chapter, Lemma lemma-quotient-by-closed, invoked at proof lines 1016-1017. Shows M^/N^ is complete and M^ -> M^/N^ is open, which lets the universal property of completion be applied to produce b.
- The universal property of the completion of a linearly topologized module. Same chapter, the discussion of completions and the author's reference to Matsumura, Theorem 8.1. Produces the canonical map b : (M/N)^ -> M^/N^.

*Source reference.* Original lemma 0ARZ: sources/stacks-formal-spaces/pinned-original-source.tex, source lines 948–948. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

*Source reference.* Original lemma 0AMT: sources/stacks-formal-spaces/pinned-original-source.tex, source lines 972–972. Pointer outside the counted span, invoked

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once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

**Lemma 0.1.** *Let $R$ be a topological ring. Let $M$ be a linearly topologized $R$-module. Let $N \subset M$ be a submodule. Assume $M$ has a countable fundamental system of neighbourhoods of $0$. Then*

(1) $0 \to N^\wedge \to M^\wedge \to (M/N)^\wedge \to 0$ *is exact,*
(2) $N^\wedge$ *is the closure of the image of* $N \to M^\wedge$,
(3) $M^\wedge \to (M/N)^\wedge$ *is open.*

*Proof.* We have $0 \to N^\wedge \to M^\wedge \to (M/N)^\wedge$ is exact and statement (2) by Lemma 0ARZ. This produces a canonical map $c : M^\wedge/N^\wedge \to (M/N)^\wedge$. The module $M^\wedge/N^\wedge$ is complete and $M^\wedge \to M^\wedge/N^\wedge$ is open by Lemma 0AMT. By the universal property of completion we obtain a canonical map $b : (M/N)^\wedge \to M^\wedge/N^\wedge$. Then $b$ and $c$ are mutually inverse as they are on a dense subset. □

## References

[Ma] Matsumura, Hideyuki. Commutative ring theory. Cambridge studies in advanced mathematics, volume 8, Cambridge University Press, 1986.